

COMP232 Individual coursework

Assignment 2

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Assignment overview and marking criteria

The coursework for COMP232 consists of two assignments, with the contribution of each as follows:

- Assignment 1 - 15%
- Assignment 2 - 15%

This document describes Assignment 2 which will be marked according to the following general criteria:

- Correctness of the arguments.
- Presence or absence of evidence on the experiments.
- Original contribution in implementation or analysis.

1 Comparison of methods for Message Authentication

This part of the assignment asks you to compare the following methods for message integrity, authentication, and non-repudiation that you have studied in Labs 5 and 6.

- Hash functions (e.g., SHA-256)
- Digital signatures
- HMAC-SHA256

Write a short, and precise, description of each of the above methods that covers the following:

1. What is it used for, and under what assumptions does it work?
2. How are integrity, authentication, and non-repudiation verified?
3. What are the advantages and disadvantages of the method?

Note: You do not have to write any code for this question. You are being asked to merely write down your understanding of the different methods that you studied in your lab sessions.

2 Generating Two Different Files with the same MD5 Hash

Using Seed Labs 2.0 installed in the University labs (or your own computer), perform **Task 1** (Section 2.1) of the **MD5 Collision Attack Lab** available at: https://seedsecuritylabs.org/Labs_20.04/Files/Crypto_MD5_Collision/Crypto_MD5_Collision.pdf

Please briefly describe your experiment and answer only Questions 1 and 3 from Task 1.

3 Key Exchange for Four parties

This part of the assignment asks you to design and implement a variant of the Diffie–Hellman key exchange protocol that allows **four parties** to establish a common shared secret. Please

- briefly describe a design for extending the two-party Diffie–Hellman protocol from the lecture to four parties
- using provided reference implementations of DH for two and three parties (Lab 4) implement your design and demonstrate that it works
- write a report on the above, providing *evidence* for your arguments.

Write a report addressing all the tasks above. Aim to be *clear*, *precise*, and *succinct*. As a general guideline, a clear and complete report for this assignment would typically be around 1,000–1,500 words, and reports within this range are entirely sufficient. This guidance is indicative only, and there is no minimum word count. Reports that are significantly longer are unlikely to be necessary and will not receive additional credit; clarity, relevance, and conciseness are valued over length. Tables, figures, code snippets, and references are *not included* in this guideline.

Note

- The marks for this assignment are distributed approximately as follows: Question 1 (30%), Question 2 (30%), and Question 3 (40%).
- The work should be formatted as a lab report. If you are unsure how to structure or write a report, see: <https://liverpool.instructure.com/courses/24284/pages/academic-writing> or the relevant Canvas page.
- Please refer to pages 3 and 4 for the detailed marking descriptor.

Submission

- The report must be submitted via **Canvas** in PDF, DOC, or DOCX format.
- **Deadline: 17:00 on 01 May 2026.**

Please note that the standard University policies set out in the [Code of Practice on Assessment \(CoPA\)](#) apply to this assignment. In particular, please refer to Section 9 for policies on plagiarism, collusion, and fabricated data, and to Section 6 for policies on late submission and associated penalties.

Grade%	Descriptor
0-9%	No evidence of the use of a suitable methodology. Virtually devoid of any evidence of knowledge or understanding of the subject. No or almost no arguments/answers.
10-19%	Little evidence of the use of a suitable methodology. Shows only the most limited and fragmentary of the subject with little or no understanding of essential principles and concepts. Work is likely to be unstructured and ill-presented. Arguments/answers are only loosely related to issues/questions or only cover a seriously inadequate part of the issues/questions.
20-29%	Fails to demonstrate an ability to solve simple, well-defined, problems of a familiar type under guidance. Serious lack of the use of a suitable methodology or flaws in its use which negatively impact on the result of the work. Very limited range of knowledge with many important gaps and omissions. Shows incomplete understanding with numerous errors of interpretation. Arguments/answers have little structure, contains serious errors, and there is no support for arguments.
30-34%	Fails to demonstrate an ability to solve simple, well-defined problems of a familiar type. Lack of the use of a suitable methodology or flaws in its use which negatively impact on the result of the work. Knowledge and understanding of the subject are fragmentary, with an insufficient number of aspects showing a very basic level of understanding and too many aspects displaying fundamental errors omissions. Arguments/answers are lacking in structure.
35-39%	Fails to demonstrate an ability to solve limited, well-defined, problems of a familiar type. Aspects of a suitable methodology evident, but flaws in its use or omissions which negatively impact on the result of the work. Knowledge and understanding of the subject are fragmentary, some aspects showing a very basic level of understanding but other aspects displaying fundamental errors. Arguments/answers are lacking in structure. There are errors and omissions and the evidence provided to support arguments is very limited.
40-49%	Demonstrates an ability to solve limited, well-defined problems of a familiar type. Most aspects of a suitable methodology evident, but minor flaws in its use or omissions with some negative impact on the result of the work. General knowledge and understanding of the subject but very limited in depth or breadth. Arguments/answers are likely to be somewhat lacking in structure. There are likely to be errors and omissions and the evidence provided to support arguments will be very limited.

Grade%	Descriptor
50-59%	Displays ability to analyse, interpret and organise information to produce coherent accounts or solve well-defined problems of some scope. Most aspects of a suitable methodology evident and used correctly, some omissions occur but without negative impact on the result of the work. Satisfactory knowledge and understanding of the essentials of the subject, with an ability to integrate information into a clear, well-structured account, but lacking in breadth or depth, or with some significant aspects omitted. Arguments/answers must be clear, although they may not be well-developed or reflect a wider appreciation of the subject. Some errors and omissions are likely to be present.
60-69%	Demonstrates ability to analyse, interpret and organise information to produce coherent accounts or solve relatively complex problems. Use of a suitable methodology evident and used correctly, with minor omissions. Good knowledge and understanding of the subject, with no major gaps or omissions, but minor gaps or omissions may occur. Arguments/answers will be clear, competently structured, and largely accurate.
70-79%	Demonstrates ability to analyse, interpret and organise information to produce coherent accounts or solve complex problems. All aspects of a suitable methodology evident and used correctly. Comprehensive knowledge and understanding of the subject together with the ability to put the work into context and to critically evaluate selected aspects of the work. Arguments/answers will be clear, competently structured, and accurate.
80-89%	Displays a level of originality and creativity and/or the ability to suggest realistic solutions to novel problems. The methodology employed is well-developed and correct. Evidence of wide reading. For level 6 this should include relevant research papers and books. Perceptive, focused treatment of all issues/questions presented in a critical and scholarly way.
90-100%	Displays an exceptional degree of originality and creativity and/or exceptional analytical and problem-solving skills. The solution must have novel aspects. The methodology employed is well-developed and correct. Shows critical understanding of current knowledge. For level 6 this should include relevant recent research papers. Perceptive, focused treatment of all issues/questions presented in a critical and scholarly way.